Overview

Model evaluation and selection determine whether model is genuinely useful or only appears good due to leakage, overfitting, or wrong metrics.

Train/Validation/Test Strategy

- **Train set:** fit model parameters.
- **Validation set:** tune hyperparameters/thresholds.
- **Test set:** final unbiased estimate.

Never tune repeatedly on test set.

Cross-Validation

k-Fold CV

Split training data into k folds; train on $k-1$ folds, validate on remaining fold; average performance.

Stratified CV

Preserves class proportions in each fold; critical for imbalanced classification.

Hyperparameter Tuning

- **Grid search:** exhaustive over predefined grid.
- **Random search:** sample configurations; often more efficient in high dimensions.
- **Bayesian optimization (high-level):** probabilistic model guides promising configurations.

Model Comparison Framework

Use common pipeline and fixed splits for fair comparison.

Consider: - Predictive performance. - Latency and memory. - Interpretability and governance constraints. - Training/inference cost.

Data Leakage and Robustness

Leakage examples: - Fitting scaler on full dataset before split. - Using future timestamps in training features.

Leakage inflates offline metrics and causes production failure.

Business Case Studies & Exceptions

Case 1: Fraud Detection Model Choice

Best AUC model had unacceptable latency for real-time scoring.

Decision: - Select slightly lower AUC model with strong recall and lower latency. - Use async secondary model for deeper review.

Case 2: Overfitted Validation Leader

Model tuned excessively on single validation split failed post-deployment.

Mitigation: - Nested or repeated CV. - Holdout challenge set by time segment.

Interview Questions & Answers

1. **Q: Why need separate test set?**
   A: To estimate generalization on unseen data after all tuning decisions are finalized.
2. **Q: What is cross-validation?**
   A: Repeated train/validate over multiple folds to reduce split variance.
3. **Q: Stratified CV use case?**
   A: Imbalanced classification where class ratios must be preserved.
4. **Q: Grid vs random search?**
   A: Grid exhaustive but expensive; random search often finds strong configs faster.
5. **Q: What is data leakage?**
   A: Train process receives info unavailable at prediction time.
6. **Q: Why model selection not only about metric score?**
   A: Real systems need latency, interpretability, and cost constraints.
7. **Q: How evaluate imbalanced data?**
   A: Use precision/recall/F1/PR-AUC and class-aware thresholding.
8. **Q: What is calibration relevance?**
   A: Threshold decisions require reliable probabilities, not only ranking.
9. **Q: Why repeated CV?**
   A: Reduces randomness from one fold partitioning.
10. **Q: Example of leakage in feature engineering?**
    A: Target encoding computed using full data including validation rows.